CHOWHAN DUTTA

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EDUCATION

SRM Institute of Science and Technology (SRMIST) Chennai, India

B.Tech in Computer Science & Engineering — AI & ML *2022 – 2026*

TECHNICAL SKILLS

- **Languages & Core:** Python, TypeScript, SQL.
- **AI/ML & Generative AI:** PyTorch, TensorFlow, Keras, ONNX Runtime, YOLOv11, Hugging Face, LangChain, OpenAI API, Ollama, RAG, OpenCV, MediaPipe.
- **Frontend, Backend & APIs:** React.js, Next.js, TypeScript, Tailwind CSS, FastAPI, Node.js, REST APIs, WebSocket.
- **Deployment & Cloud:** Docker, Vercel, Render, Firebase, Hugging Face Spaces.

PROJECTS

ShieldAI — Real-Time Weapon Detection System

Python · FastAPI · YOLOv11 · ONNX Runtime · OpenCV · WebSocket · Docker · HuggingFace

- Fine-tuned YOLOv11s on multi-source Roboflow datasets across 4 weapon classes (handgun, long gun, knife, explosive) — trained 120 epochs on Google Colab T4 GPU
- Optimized inference using ONNX Runtime for CPU deployment — achieving low-latency real-time detection via WebSocket streaming from webcam and CCTV/IP camera streams
- Implemented enhanced knife detection using quadrant-crop scanning at 1280px, significantly improving recall for small bladed objects at distance
- Containerized with Docker and deployed live on Hugging Face Spaces with 100% local processing — zero external API calls

RAGs AI — Multi-Format Document Q&A System

Python · FastAPI · LangChain · React.js · Vercel · Render

- Built end-to-end RAG pipeline from scratch supporting PDF, Word, and CSV uploads — users query documents in plain English and receive grounded, context-aware answers in real time
- Designed and optimized semantic chunking and vector retrieval logic — reducing irrelevant context retrieval and improving answer accuracy
- Deployed full-stack with FastAPI backend on Render and React frontend on Vercel — production-ready with live demo

Local SLM Benchmark — AI Model Evaluation Tool

Python · FastAPI · Ollama · phi3 · llama3.2 · gemma2

- Built a full-stack benchmarking tool to compare Small Language Models (phi3, llama3.2, gemma2) running locally via Ollama — featuring real-time metrics: tokens/sec, latency, response duration
- Designed 30-prompt evaluation suite across 10 categories (reasoning, coding, math, creativity, translation) with auto-generated technical reports, rankings, and per-category charts
- Implemented Temperature Lab to visualize how output quality shifts across 0.0–1.0 temperature range, with structured JSON output mode and auto-retry on validation failure

Handwritten Character Recognition — CNN + GAN

Python · TensorFlow · Keras · GAN · OpenCV · NumPy

- Designed an OCR system combining CNN for spatial feature extraction with GANs for synthetic data augmentation — improving model robustness across diverse handwriting styles
- Built a 5-stage pipeline: image preprocessing, CNN classification, GAN-based augmentation, context-aware integration layer, and final output generation